	TOTAL FOR QUESTION 6	**11**
7(a)	Half-life is the <u>average</u> time taken for half the original <u>number</u> of radioactive <u>nuclei</u> to decay.	**B1**
7(b)(i)	$A = \lambda N$ $1.6 \times 10^{14} = \dfrac{ln\,2}{3.6 \times 24 \times 3600} N$ [M1] $N = 7.18 \times 10^{19}$ Mass = $7.18 \times 10^{19} \times 224 \times 1.66 \times 10^{-27}$ [M1] $= 2.67 \times 10^{-5}$ kg $= 27$ mg [A0]	**M1** **M1** **A0**

Qns	Answer	Marks
7(b)(ii)	Consider activity of 5.0mg of the fresh sample $A_o = \frac{5.0}{2.67} \times 1.6 \times 10^{14}$ $= 2.996 \times 10^{13}$ Bq	M1
	$\frac{A}{A_o} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$ $\frac{2.1 \times 10^{10}}{2.996 \times 10^{13}} = \left(\frac{1}{2}\right)^{\frac{t}{3.6}}$	M1
	$t = 38$ days	A1
	TOTAL FOR QUESTION 7	**6**